

# Regression Discontinuity

BUSS975: Causal Inference in Financial Research

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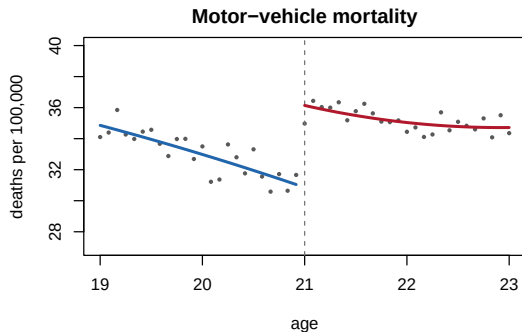
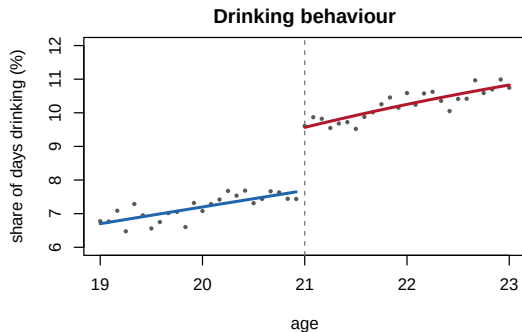
Korea University

## 6.1 Bird's-Eye View of RD

# Where we left off

- Lecture 3 : **Matching** needs common support; **regression** needs a correctly specified outcome model and will extrapolate where the data are silent.
- In regression discontinuity, at any value of the running variable, you observe treated units **or** control units. Overlap is exactly zero.
- So extrapolation is the entire design. The question becomes: *how short a distance can we extrapolate?*
- The answer is what makes RD a credible observational design we have.

# The intuition: alcohol consumption on mortality - drinking at 21



Drinking rises at 21 because that is when it becomes legal. Motor-vehicle mortality falls steadily with age, then jumps back up at exactly the same point.

# What makes it an RD figure

Three elements, and every RD figure has all three:

1. **The dots.** Average outcomes within narrow bins of the running variable — here, one dot per month of age.
2. **The lines.** Fitted values from a regression, estimated **separately on each side** of the cutoff. The curvature comes from a polynomial in age.
3. **The break.** That gap is the estimate.

# The jargon

- **Running variable** ( $X$ ): the score that assigns treatment. Ideally continuous; a fine enough grid will do.
- **Cutoff** or **threshold** ( $c_0$ ): the known value where assignment changes.
- **Trends**: how the *outcome* moves with the running variable — **not** time trends. In the mortality figure, deaths fall with age; that is an outcome trend.
- **Discontinuity** or **jump**: the break in the conditional mean at  $c_0$ .

# Where do running variables come from?

- They are everywhere, because **humans write rules**, and rules need thresholds.
- **Governments**: the drinking age at 21, adult criminal liability at 18.
- **Firms and markets**: credit-score cutoffs in lending, index membership rules, EPS targets in compensation contracts, a 50% vote share.
- Finding one requires **institutional knowledge**, not econometrics. Knowing the rule exists, and exactly how it is applied, is the hard part.

## And RD needs a great deal of data

- The estimate comes from observations **near the cutoff**.
- So the binding constraint is the number of rows in a neighbourhood of one point.
- Strong trends make this worse: the more steeply the outcome moves with  $X$ , the more data you need to separate the jump from the slope.
- This is why RD studies are overwhelmingly built on **administrative records** — birth registries, school records, court records, credit bureaus.
- It is also why RD in finance is often underpowered.

## 6.2 Identification Assumptions

# Sharp and fuzzy

|                              | Sharp                           | Fuzzy                    |
|------------------------------|---------------------------------|--------------------------|
| $\Pr(D = 1 \mid X)$ at $c_0$ | jumps $0 \rightarrow 1$         | jumps, but not to 1      |
| Treatment is                 | a deterministic function of $X$ | influenced by $X$        |
| The cutoff is                | the assignment rule             | an <i>instrument</i>     |
| Example                      | Medicare at 65                  | Israeli class-size rules |

- We do the **sharp** design here. The fuzzy design is an instrumental-variables problem, so it waits for the IV lecture.
- Many finance applications are fuzzy. A shareholder proposal that passes with 51% is not always implemented.

# The sharp design

In a sharp RD, treatment status is a **deterministic, discontinuous** function of the running variable:

$$D_i = \begin{cases} 1 & \text{if } X_i \geq c_0 \\ 0 & \text{if } X_i < c_0 \end{cases}$$

- Know  $X_i$  and you know  $D_i$  with certainty. Know someone's age and their country, and you know whether they may legally drink.
- Contrast the fuzzy design, where crossing  $c_0$  raises  $\Pr(D = 1)$  without driving it to one.

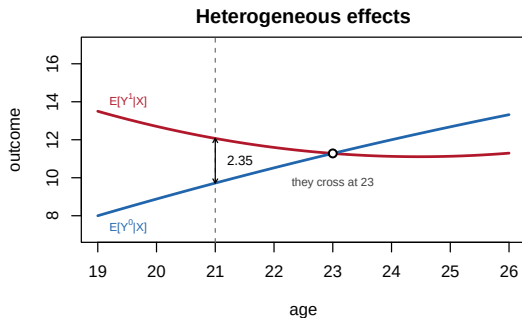
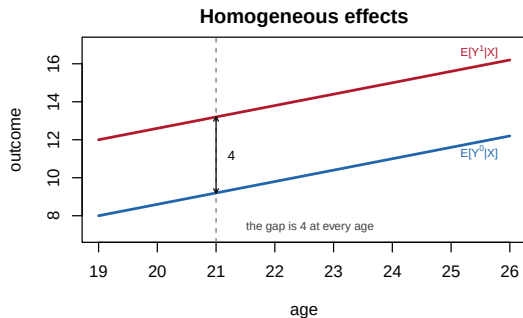
## The key feature: no common support

- This is a selection-on-observables design: condition on  $X$  and treatment is fully determined.
- But the other assumption of Lecture 3 fails completely. At any value of  $X$  you see treated units or controls, never both:

$$e(X) = \Pr(D = 1 \mid X) \in \{0, 1\} \quad \text{everywhere.}$$

- No 19-year-old may legally drink; every 21-year-old may. There is no overlap to exploit.
- **Every tool from last week is unusable.** You cannot match, stratify, or weight — no unit has a counterpart at its own value of  $X$ .
- What replaces overlap is a **smoothness** assumption, plus a limiting argument at a single point.

# The target parameter



Left: parallel potential outcomes, so the vertical gap is the same at every age and the number at the cutoff describes everyone.

Right: the functions curve and **cross at 23** — the effect is 5.5 at age 19, 2.35 at the cutoff, zero at 23, and *negative* beyond. RD returns the gap **at the dashed line** in both panels.

# The estimand is a limit

$$\tau = \underbrace{\lim_{x \downarrow c_0} E[Y_i | X_i = x]}_{\text{from the right}} - \underbrace{\lim_{x \uparrow c_0} E[Y_i | X_i = x]}_{\text{from the left}} = E[Y_i^1 - Y_i^0 | X_i = c_0]$$

- It is only *in the limit* that overlap is restored. Approaching  $c_0$  from either side, the two groups become arbitrarily similar in  $X$ .
- This is an *ATE*, but for the subpopulation  $\{X = c_0\}$  — which is why it is often called a **local** average treatment effect.
- Data from a wide range of  $X$  are used only to estimate those two limits precisely. The estimand still lives at  $c_0$ .

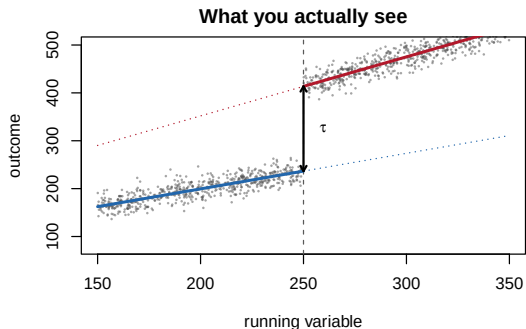
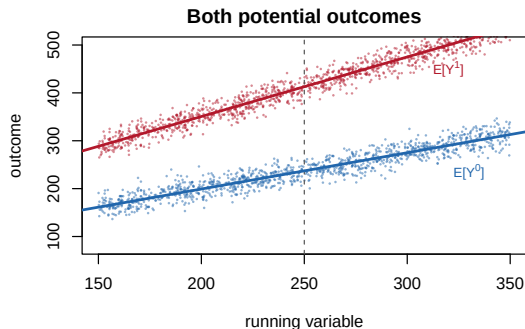
# Smoothness: the assumption that does the work

- **Hahn, Todd and Van der Klaauw (2001)** proved the effect at the cutoff is identified if the conditional expectations of **both** potential outcomes are continuous at  $c_0$ :

$$E[Y_i^0 \mid X_i = x] \quad \text{and} \quad E[Y_i^1 \mid X_i = x] \quad \text{are continuous at } x = c_0.$$

- A jump is evidence that *something* happened, and the design's job is to make treatment the only candidate.

# Smoothness, in a simulation



$Y^1 = 100 + 1.25X + \varepsilon$  and  $Y^0 = 50 + 0.75X + \varepsilon$ . Neither jumps at  $X = 250$  — **nothing happens at the cutoff** in the potential outcomes. The *observed* outcome jumps anyway, by exactly the vertical distance between the two functions there. That distance is  $\tau$ .

# The price of the design

- **You get** an effect that is credible, visible, and hard to fake — identified by a rule you can read in a statute or a contract.
- **You pay** by learning about one point. If the policy question is “what would this do to everyone?”, RD does not answer it.
- The gap between what RD identifies and what a policymaker wants is often large. A study of a drinking age of 21 says nothing about a drinking age of 18.
- State the estimand as what it is: *“the average effect for units at the threshold.”* Not “the effect.”

## 6.3 Estimation

# Two steps

**Step 1. Recentre the running variable** at the cutoff:

$$\tilde{X}_i = X_i - c_0.$$

**Step 2. Regress the outcome on treatment, the running variable, and their interaction:**

$$Y_i = \alpha + \tau D_i + \beta_1 \tilde{X}_i + \beta_2 D_i \cdot \tilde{X}_i + \varepsilon_i.$$

- $\beta_1$  is the slope **below** the cutoff,  $\beta_1 + \beta_2$  the slope **above**, and  $\hat{\tau}$  estimates the jump.
- The interaction lets the two expected potential-outcome functions have different slopes. Drop it and any difference in trend is loaded onto  $\hat{\tau}$ .

## Where the specification comes from

Model each potential outcome as a  $p$ th-order polynomial in the centred running variable:

$$E[Y^0 | X] = \mu + \beta_{01}\tilde{X} + \cdots + \beta_{0p}\tilde{X}^p, \quad E[Y^1 | X] = \mu + \tau + \beta_{11}\tilde{X} + \cdots + \beta_{1p}\tilde{X}^p$$

Apply the **switching equation**,  $Y_i = D_i Y_i^1 + (1 - D_i) Y_i^0 = Y_i^0 + D_i(Y_i^1 - Y_i^0)$ , and take expectations given  $X$ :

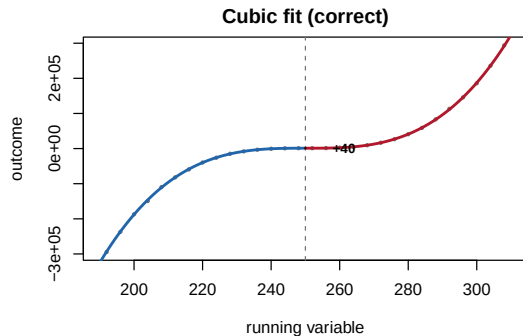
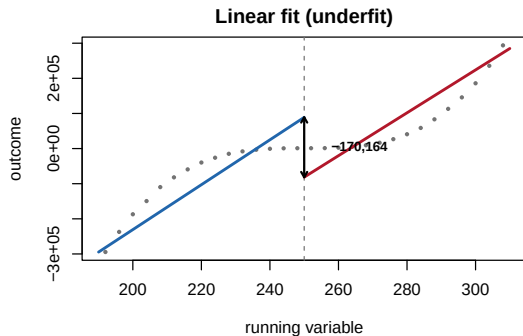
$$E[Y | X] = E[Y^0 | X] + D(E[Y^1 | X] - E[Y^0 | X])$$

which is the regression you actually run:

$$Y_i = \mu + \beta_{01}\tilde{X}_i + \cdots + \beta_{0p}\tilde{X}_i^p + \tau D_i + \gamma_1 D_i \tilde{X}_i + \cdots + \gamma_p D_i \tilde{X}_i^p + \varepsilon_i$$

- with  $\gamma_k = \beta_{1k} - \beta_{0k}$ . Interacting  $D$  with **every** polynomial term is what lets the two sides take genuinely different shapes.
- The treatment effect **at the cutoff** is  $\tau$ .

# Get the shape wrong and the answer is nonsense



The truth is a **cubic** in the running variable and the treatment effect is exactly **zero** — the two potential outcomes are identical, so they touch everywhere. Straight lines cannot bend where the data bend, so they arrive at the cutoff at the wrong heights and manufacture a jump out of nothing.

## The damage, quantified

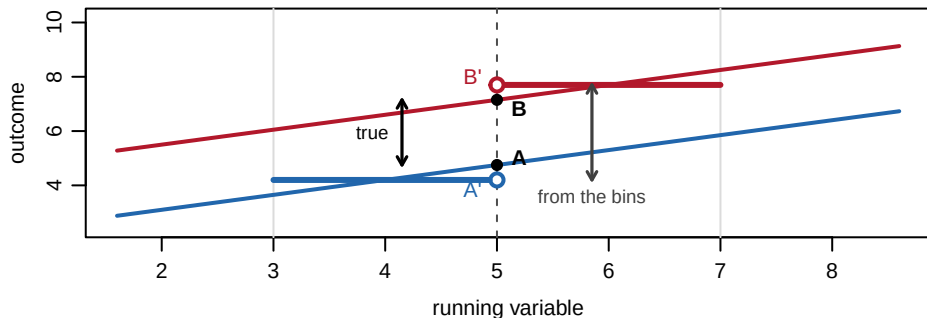
| Fitted polynomial | $X$      | $X, X^2$ | $X, X^2, X^3$ |
|-------------------|----------|----------|---------------|
| $\hat{\tau}$      | -170,164 | 59,136   | 40            |

- **The true effect is zero.** Underfit by two orders and you get about  $-170,000$ ; by one, about  $+59,000$ . Get the order right and the estimate collapses into noise.
- These are not merely biased — they are wrong by *five orders of magnitude*, and they would be reported as **highly statistically significant**.
- The mechanism is the extrapolation.  $\hat{\tau}$  is the gap between two fitted values **at a boundary**, and boundary predictions are exactly where a misspecified polynomial errs most.

## But do not solve it with more polynomials

- The obvious fix — keep raising  $p$  — is the wrong one. **Gelman and Imbens (2019)** give three reasons not to go above quadratic:
  1. High-order polynomials put **large, erratic weights** on observations far from the cutoff, so distant data drive an estimate that is supposed to be local.
  2. Estimates are **highly sensitive** to the order chosen, with no principled way to pick it.
  3. The resulting confidence intervals have **poor coverage**.
- Their recommendation, now standard: **local linear or local quadratic**, fitted only near the cutoff.
- Which reframes the problem. Instead of “what shape fits the whole range?”, ask “how small a neighbourhood can I afford, and fit a line inside it.”

# The boundary problem with “rectangular kernel”



The true effect is  $\overline{AB} = 2.4$ . But a bin mean sits at the bin's **centre**, not at its edge, so the left limit is read off at  $A'$  and the right at  $B'$ : the estimate is  $\overline{A'B'} = 3.5$ , too large by 46%. **The trends cause the bias** — flatten the two lines and  $A' \rightarrow A$ ,  $B' \rightarrow B$ .

# Kernel and bandwidth

- The standard fix (Hahn, Todd and Van der Klaauw): fit a **line** inside the window rather than a mean, and let it extrapolate the short distance to the boundary. That is local linear regression, and it substantially reduces the bias.

$$(\hat{a}, \hat{b}) = \arg \min_{a,b} \sum_{i=1}^n (Y_i - a - b(X_i - c_0))^2 K\left(\frac{X_i - c_0}{h}\right) \mathbf{1}\{X_i > c_0\}$$

- Think of kernel regression as a weighted regression restricted to a window.
  - ▶ The **kernel**  $K(\cdot)$  is the *shape* of the window — how weight declines with distance from the cutoff.
  - ▶ The **bandwidth**  $h$  is the *width* of the window — how far from the cutoff you are willing to look.

## Three choices: $h$ , $K(\cdot)$ , and $p$

- **The bandwidth  $h$ .** Smaller: fewer observations, less bias, more variance. Larger: more data, less variance, more bias if the included units depart from the true function. This choice matters most.
- **The kernel  $K(\cdot)$ .** Next slide.
- **The polynomial order  $p$ .**  $p = 0$  is a difference in means — intuitive, but it *is* the boundary problem.  $p > 1$  addresses it but risks overfitting and instability. **Local linear ( $p = 1$ ) is generally preferred.**
- Calonico, Cattaneo and Titiunik (2014) recommend the **triangular kernel with an MSE-optimal bandwidth**: maximum weight at the cutoff, zero at the window's edge.
- There is no universally correct answer. Report what you chose, and show the answer does not turn on it.

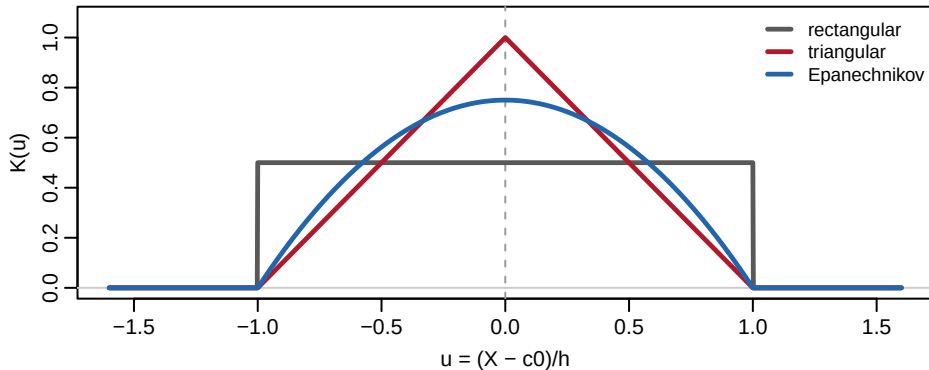
# The three kernels, written out

Let  $u = (X_i - c_0)/h$ , so that  $|u| \leq 1$  means “inside the window”.

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|                       |                                                        |
|-----------------------|--------------------------------------------------------|
| Rectangular (uniform) | $K(u) = \frac{1}{2} \mathbf{1}\{ u  \leq 1\}$          |
| Triangular            | $K(u) = (1 -  u ) \mathbf{1}\{ u  \leq 1\}$            |
| Epanechnikov          | $K(u) = \frac{3}{4}(1 - u^2) \mathbf{1}\{ u  \leq 1\}$ |

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# The recipe

1. Choose a polynomial order  $p$  and a kernel  $K(\cdot)$ .
2. Select a bandwidth  $h$ .
3. **Above the cutoff**, run weighted least squares of  $Y$  on a constant and  $p$  terms in  $(X_i - c_0)$ , with weights  $K((X_i - c_0)/h)$ . The intercept estimates  $\mu_+ = \lim_{x \downarrow c_0} E[Y | X = x]$ :

$$\hat{Y}_i = \hat{\mu}_+ + \hat{\mu}_{+,1}(X_i - c_0) + \cdots + \hat{\mu}_{+,p}(X_i - c_0)^p$$

4. **Repeat below the cutoff** for  $\hat{\mu}_-$ .
5. The sharp RD point estimate is the difference of the two intercepts:

$$\hat{\tau} = \hat{\mu}_+ - \hat{\mu}_-.$$

- In practice this is one line: `rdrobust(y, x, c = 0)` in R, `rdrobust y x, c(0)` in Stata.

## What “weighted least squares” means in step 3

Give unit  $i$  the weight  $w_i = K((X_i - c_0)/h)$  and minimise the **weighted** sum of squares:

$$\hat{\beta} = \arg \min_{\beta} \sum_{i=1}^n w_i (Y_i - x_i' \beta)^2, \quad x_i = (1, \tilde{X}_i, \tilde{X}_i^2, \dots, \tilde{X}_i^p)', \quad \tilde{X}_i = X_i - c_0$$

with the closed form  $\hat{\beta} = (X'WX)^{-1}X'WY$ , where  $W = \text{diag}(w_1, \dots, w_n)$ .

- Units with  $|\tilde{X}_i| > h$  get  $w_i = 0$  and contribute nothing to either  $X'WX$  or  $X'WY$ .
- **Rectangular kernel**  $\Rightarrow$  **plain OLS on the subsample**, because every in-window weight is then equal.

# Choosing the bandwidth

Bias grows with  $h$ , variance falls with it. Writing each as a function of  $h$  and minimising:

$$MSE(h) \approx \underbrace{B^2 h^{2(p+1)}}_{\text{bias}^2} + \underbrace{\frac{V}{nh}}_{\text{variance}} \quad \Rightarrow \quad h_{MSE} \propto \underbrace{V^{\frac{1}{2p+3}}}_{\text{noise}} \underbrace{B^{-\frac{2}{2p+3}}}_{\text{curvature}} \underbrace{n^{-\frac{1}{2p+3}}}_{\text{sample size}}$$

- **Noise  $V$ : positive.** Noisier data  $\Rightarrow$  a **wider** window, to average the noise away. If  $V \rightarrow 0$ , then  $h \rightarrow 0$  — use only points at the cutoff.<sup>1</sup>
- **Curvature  $B$ : negative.** More curvature  $\Rightarrow$  a **narrower** window. If  $B \rightarrow 0$  (a truly linear CEF),  $h \rightarrow \infty$  — just run OLS on everything.
- **Sample size  $n$ : negative.** More data  $\Rightarrow$  a **narrower** window.

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<sup>1</sup>The Remix states the  $n$  and noise directions the other way round.

# Why the obvious confidence interval is wrong

- The local polynomial is a **nonparametric approximation** to an unknown function. It does not fit perfectly inside the window, so  $\hat{\tau}$  carries an approximation bias  $B$  as well as sampling noise. Honest inference needs

$$CI = [(\hat{\tau} - B) \pm 1.96\sqrt{V}].$$

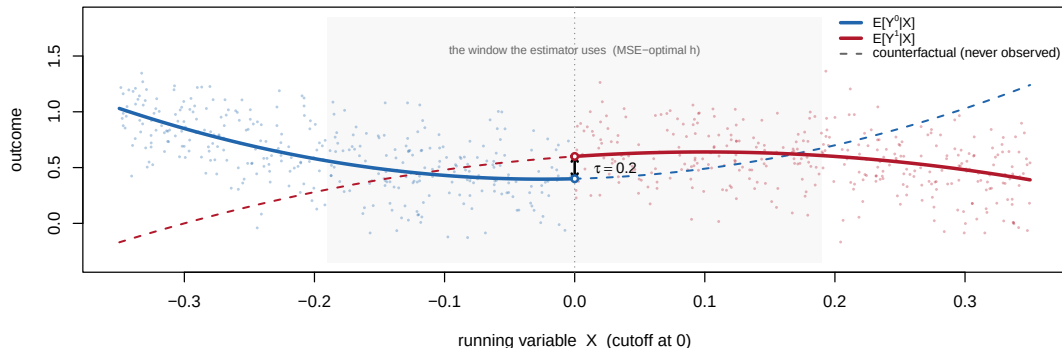
- The conventional interval,  $CI_{us} = [\hat{\tau} \pm 1.96\sqrt{\hat{V}}]$ , simply **assumes**  $B = 0$  — it centres the interval on a biased point estimate.
- The consequence is **over-rejection** of the null of no effect. Cattaneo, Jansson and Ma (2019) discourage reporting these.

# Three intervals

| Interval                          | Centred at             | Standard error        | Verdict                                |
|-----------------------------------|------------------------|-----------------------|----------------------------------------|
| Conventional, $CI_{us}$           | $\hat{\tau}$           | $\sqrt{\hat{V}}$      | do not use                             |
| Bias-corrected, $CI_{bc}$         | $\hat{\tau} - \hat{B}$ | $\sqrt{\hat{V}}$      | better, but ignores noise in $\hat{B}$ |
| Robust bias-corrected, $CI_{rbc}$ | $\hat{\tau} - \hat{B}$ | $\sqrt{\hat{V}_{bc}}$ | <b>report this</b>                     |

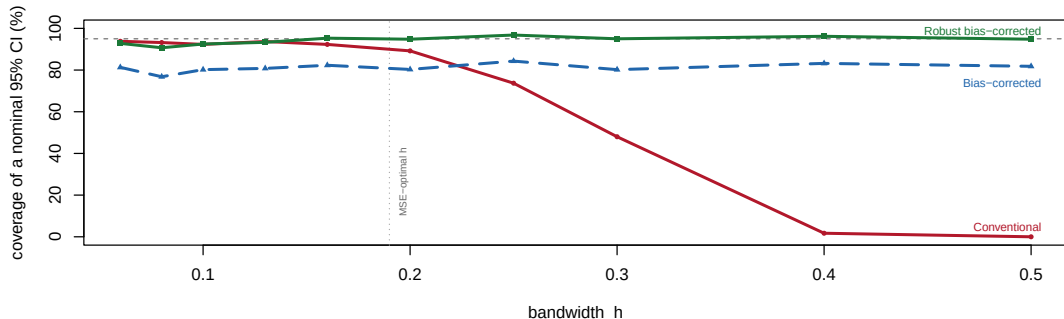
- Bias correction subtracts an estimate of  $B$ . But  $\hat{B}$  is itself estimated, and pretending otherwise distorts coverage in finite samples.
- **Robust** bias correction folds the variability of  $\hat{B}$  into the variance term. Intervals get somewhat wider for the same bandwidth — that is the honest price of the correction.
- This is what `rdrobust` reports by default. Report the robust, and say that you did.

# Simulation



- $X_i \sim U(-1, 1)$ ,  $n = 2,000$ , noise  $N(0, 0.25^2)$ , with  $m_0(x) = 0.40 + 0.30x + 6.0x^2$  below the cutoff and  $m_1(x) = 0.60 + 0.80x - 4.0x^2$  above it.
- $\tau = 0.20$  is the vertical gap between the two open circles at  $X = 0$  — and it is *known*, which is the only reason we can ask whether an interval covers it.
- **The truth is quadratic; the estimator fits a straight line on each side inside the shaded window.** That deliberate mismatch is the approximation bias  $B$ , and it grows as the window widens.

# A coverage experiment

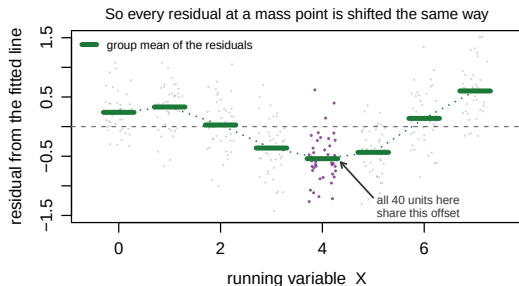
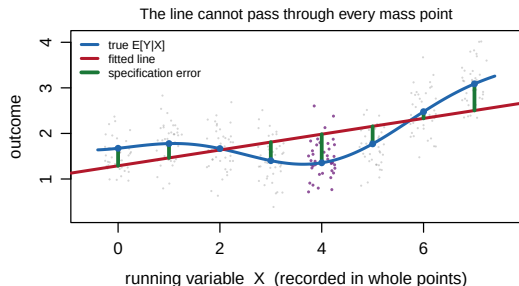


2,000 samples drawn from the previous slide's design, at each of ten fixed bandwidths, with a local **linear** fit on each side. For each sample we build all three intervals and ask: **does it contain the true  $\tau = 0.20$ ?** A 95% interval should say yes 95% of the time.

# Why RD researchers clustered on the running variable

- **Discrete running variables** create mass points where polynomial smoothing leaves shared **specification error**.
- **Lee & Card (2008)** treated this error as a group shock, popularizing clustering standard errors on unique values of  $X$ .
- This widely adopted practice is now recognized as methodologically flawed.

# What “the same specification error” means



Let  $m(x) = \mathbb{E}[Y | X = x]$  and  $m^*(x)$  be the linear approximation:

$$\underbrace{Y_i - m^*(X_i)}_{\text{Total residual}} = \underbrace{m(X_i) - m^*(X_i)}_{\text{Specification error } a_{g(i)}} + \underbrace{Y_i - m(X_i)}_{\text{Noise } u_i}$$

**Shared error:** Specification error depends solely on  $X_i$ , making it identical for all observations at a given mass point. **Group shock structure:** Because mass-point residuals do not average to zero, they share a systematic shock  $\varepsilon_i = a_{g(i)} + u_i$ , where  $g$  = the distinct value of  $X$ , which forms the basis for Lee–Card clustering.

## A short detour: what clustering a standard error does

- OLS standard errors assume the errors  $\varepsilon_i$  are **independent across observations**. Each observation is a fresh piece of information.
- Suppose instead the data arrive in **groups**, and everyone in a group shares a common shock: students in a classroom share a teacher, firms in an industry share a shock. Then residuals inside a group move together.
- Independent draws are being double-counted. Your **effective** sample size is closer to the *number of groups* than to the number of observations, and conventional standard errors are too small.
- A **cluster-robust** standard error allows arbitrary correlation *within* each group while assuming independence *across* groups. It normally makes standard errors **larger**.
- The formula's justification is asymptotic **in the number of clusters**  $G$ . With few clusters it is unreliable, and there is no warning message.

# Why RD Clustering Fails

| Bandwidth<br>( $h$ ) | $N$    | Clusters ( $G$ ) | Robust SE | Clustered<br>SE | Ratio |
|----------------------|--------|------------------|-----------|-----------------|-------|
| 0.050                | 90,074 | 101              | 0.00450   | 0.00333         | 0.74  |
| 0.020                | 38,349 | 41               | 0.00657   | 0.00421         | 0.64  |
| 0.010                | 20,154 | 21               | 0.00911   | 0.00476         | 0.52  |
| 0.005                | 10,567 | 11               | 0.01303   | 0.00740         | 0.57  |

- **Counterintuitive Shrinking:** In Hansen's BAC data, clustering *shrinks* standard errors by up to 48%, inflating Type I error rates (Kolesár & Rothe).
- **Cluster Deficit:** Narrowing bandwidth  $h$  severely depletes cluster count  $G$  (down to 11), violating cluster-asymptotic assumptions.
- **Deterministic Error:** Specification error is a fixed function of  $x$ , not a random group shock; clustering incorrectly treats deterministic bias as sampling variability.

# Recommended Alternatives

- **Heteroskedasticity-Robust SEs:** Default in `rdrobust`; avoids modeling deterministic specification error as random shocks.
- **Honest Confidence Intervals:** `RDHonest`, Armstrong & Kolesár
- **Local Randomization Inference:** (`rdlocrand`, Cattaneo et al.) Ideal for sparse/coarse data; treats a narrow window around the cutoff as a randomized experiment.
- **Evaluating Legacy Papers:** Check whether older RD results clustered on  $X$  remain statistically significant under robust standard errors.

## 6.4 Challenges to Identification

# Five ways continuity fails

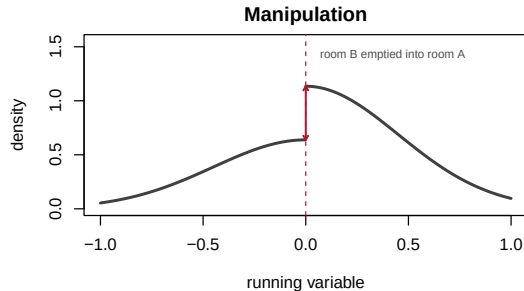
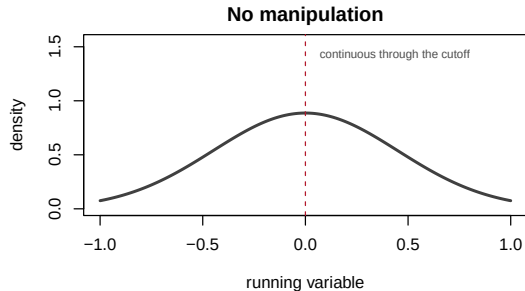
Continuity is the whole design. It breaks when:

1. The **assignment rule is known** in advance;
  2. Agents are **interested** in adjusting their score;
  3. Agents **have time** to adjust;
  4. The **cutoff is endogenous** to factors that cause potential outcomes to shift;
  5. There is **nonrandom heaping** along the running variable.
- *“Does anything else change at the cutoff?”*: You answer it by knowing the institution.

# Sorting, and the two rooms

- Two rooms of patients. Room A gets the life-saving treatment; room B gets nothing. If you are in room B, you walk across the hall — unless crossing is impossible.
- If everyone crosses, A is crowded and B is empty. That imbalance is the **observable footprint** of sorting: motives are invisible, but headcounts are not.
- Formally: with a desirable treatment assigned by  $X \geq c_0$ , agents who *can* choose  $X$  just above  $c_0$ . If that ability correlates with potential outcomes, the comparison at the cutoff is contaminated.
- A **testable prediction**: a discontinuity in the *density* of the running variable.

# The McCrary density test (2008)



- Under the null of no manipulation the density of  $X$  is **continuous** at  $c_0$ ; under the alternative it bunches on the treated side.
- Mechanically: partition the running variable into bins, count observations per bin, treat the counts as an outcome, fit a local linear regression on each side, and test whether the limits agree.
- Use `rddensity` (Cattaneo, Jansson and Ma)

# Running the test: rddensity on real data

```
library(rddensity)
dwi <- read.csv("data/hansen_dwi.csv")
bac <- dwi$bac1[dwi$bac1 > 0]

out <- rddensity(bac, c = 0.08)      # X and the cutoff.
summary(out)                        # T = -0.14, p = 0.89

rdplotdensity(out, bac)             # the picture that belongs in the paper
```

- Take the **running variable only**. You are testing the distribution of  $X$ , nothing else.
- Read the **robust bias-corrected**  $T$  statistic and its  $p$ -value.
- Always run `rdplotdensity` as well. Examine the entire distribution.

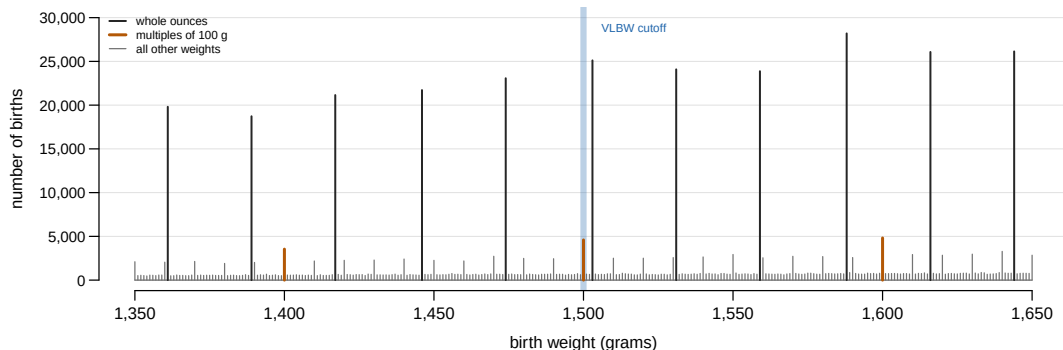
# Covariate balance and placebo cutoffs

- **Covariate balance.** Pre-treatment characteristics cannot be caused by treatment, so they must not jump at  $c_0$ . Run the same RD with each covariate as the outcome and find nothing.
- A jump in a covariate is serious. It means something *other than* treatment changes at the cutoff — and if an observed characteristic jumps, unobserved ones probably do too.
- **Placebo cutoffs.** Imbens and Lemieux (2008): take one side of the discontinuity, use the median of the running variable there as a fake cutoff  $c'_0$ , and test for a jump in the outcome. You do not want to find anything.
- The identifying assumptions cannot be tested directly. What these do is supply **indirect evidence** that may be persuasive — placebo logic, looking where there should be no effect.

# Nonrandom heaping: a design that passed its tests

- **Almond, Doyle, Kowalski and Williams (2010)** wanted the returns to medical care — which OLS cannot give you, because doctors assign care by expected outcome. The Perfect Doctor problem again.
- Their cutoff: babies below 1,500g are “**very low birth weight**” and receive heightened attention. Birth weight is the running variable, 1,500g the threshold.
- One-year mortality came out about **one percentage point lower** just below it, against a mean of 5.5%.
- They ran the **McCrary test at 1,500g and found nothing** — density continuous, no evidence of manipulation.
- Everything about this is the textbook procedure.

# Then look at the whole distribution



**Barreca, Guldi, Lindo and Waddell (2011)** looked at the *shape* of the recorded distribution rather than its density at one point. Enormous piles at whole **ounces** ( $\approx 28.35\text{g}$  apart) and smaller ones at every round 100g. Nature does not pile babies at 1,417g — and note where the rule falls: **1,500 is itself a heap point.**

# Why heaping fakes an effect at *every* cutoff

- **Heaped babies are different babies.** Rounding is not random — it happens in rushed deliveries and less-equipped hospitals. They have less-educated mothers, less prenatal care, and **higher mortality** than their unheaped neighbours.
- **The heap lands on the control side.** VLBW means  $< 1,500\text{g}$ , so a baby recorded at *exactly*  $1,500\text{g}$  is untreated. The pile sits in the control group at the boundary and drags its limit **up**, which makes being *below* the cutoff look protective.
- The same spurious benefit is waiting at  $1,400\text{g}$ ,  $1,600\text{g}$ ,  $1,700\text{g}$ .
- So what jumps is not the care these babies receive but **who is recorded there**: composition, not treatment.

# The donut-hole RD

- If units *at* the cutoff are suspect — heaped, rounded, or manipulated — but units nearby are fine, drop the suspect ones and re-estimate. That is the **donut hole**.
- In Almond et al.'s data, dropping about 2% of observations cut the estimated one-year mortality effect roughly **in half**.
- You lose observations you often cannot spare, and the estimand changes: it is now local to a population with a hole in its middle — harder to describe, and further still from anything a policymaker asked about.
- Heaping is not fatal. But report both, with and without the donut, and let the reader see the difference.

## 6.6 DUI and Recidivism (*replication exercise*)

# The question: can we see deterrence?

- **Becker (1968)**. Crime is deterred through two channels: raising the **probability of arrest**, or raising the **punishment** conditional on conviction.
- **Why aggregate data cannot answer this**. Harsher penalties do two things at once. They may **deter**, as offenders rationally reoffend less; they certainly **incapacitate**, by removing offenders from the population at risk.
- **Hansen (2015)** Imagine flipping a coin over two near-identical groups who have all drunk and driven, and punishing one group harder. If deterrence is real, that group reoffends less.

# The design

**Hansen (2015).** A driver stopped on suspicion in Washington State blows into a breathalyzer. At 0.08 or above they are charged with DUI; 0.15 marks “aggravated” DUI. Use 0.08.

|                    | 1st offence |            | 2nd offence |            |
|--------------------|-------------|------------|-------------|------------|
|                    | DUI         | Agg. DUI   | DUI         | Agg. DUI   |
| Minimum fine       | \$865.50    | \$1,120.50 | \$1,120.50  | \$1,545.50 |
| Minimum jail time  | 24 hours    | 48 hours   | 30 days     | 45 days    |
| Licence suspension | 90 days     | 365 days   | 2 years     | 900 days   |

- Treatment is assigned by  $\mathbf{1}\{\text{BAC} \geq 0.08\}$ , so BAC is the running variable. Compare drivers who blew 0.079 with those who blew 0.081: if those just above reoffend less, the escalated punishment deterred them.
- **The outcome is recidivism** — the same person stopped again above 0.08. Measurable only because the administrative records carry persistent personal identifiers.
- **Manipulation is implausible.**
- Your task: replicate it!

# A checklist for an RD paper

1. **Name the rule.** Cite the statute, policy document or contract. Say who assigns the score and who can see it.
2. **Plot the density** of the running variable and run a formal test. Look for heaping, not just a jump.
3. **Plot the outcome** in bins, with fits on each side. If the effect is not visible, be sceptical of the regression that finds it.
4. **Show covariate balance**, in a table and a figure, using the same specification as the main result.
5. **Report the main estimate** with a triangular kernel, an MSE-optimal bandwidth, and **robust bias-corrected** intervals.
6. **Vary the bandwidth** and plot the estimates. Vary the polynomial order too, staying at or below quadratic.
7. **Run placebo cutoffs** where nothing should happen.
8. **State the estimand** as a local effect at the threshold, and resist calling it “the” effect.
9. If the running variable is heaped, consider a **donut-hole** estimate too.

## RD in Finance

## Where the cutoffs are

- Finance is full of rules with thresholds, which is why RD has spread quickly in the field.

| Paper                   | Running variable          | Cutoff           | Design    |
|-------------------------|---------------------------|------------------|-----------|
| Keys et al. (2010)      | FICO credit score         | 620              | sharp-ish |
| Malenko and Shen (2016) | ISS voting-policy score   | policy threshold | sharp     |
| Almeida et al. (2016)   | EPS relative to consensus | 0                | sharp     |
| Cuñat et al. (2012)     | shareholder vote share    | 50%              | fuzzy     |

- Index membership rules, covenant thresholds, credit ratings at the investment-grade boundary, regulatory size thresholds — all candidates.
- Ask how plausible **no manipulation** is: in finance the agents are sophisticated, well advised, and highly motivated.

## 6.8 Concluding Remarks

# What the design buys

- **Treatment assignment is known**, not assumed. You can read the rule in a statute or a contract, which is what makes RD the most credible design among observational methods.
- **Unconfoundedness is free**. Conditional on the running variable, treatment is deterministic — so the assumption that carried all the weight in Lecture 3 is satisfied by construction.
- **The identifying assumption is about smoothness**, and smoothness has testable implications: a continuous density, balanced covariates, no effects at placebo cutoffs.
- **The evidence is visual**. A reader can see the design work, or fail, before any regression appears.
- **And it seems harder to game**. Brodeur, Cook and Heyes (2020) find RD studies show less sign of  $p$ -hacking than other observational designs — plausibly because so much raw data is on display, the regressions are simple, and there are few researcher choices with real power to squeeze water from a rock.

# What it costs

- **The estimand is local**, at one point of one variable. It is not the *ATE* and generally cannot be extrapolated without further assumptions.
- **Overlap is zero**, so everything rests on extrapolating to a boundary — and boundary estimation is exactly where misspecification does the most damage.
- **The design is data-hungry**, because it estimates effects at the boundary of a cutoff. In finance that often means a few dozen observations doing the work.
- **Diagnostics are necessary, not sufficient.** Almond et al. passed the density test, and the donut hole still moved their answer by half.
- Next: the fuzzy design, where the cutoff shifts the *probability* of treatment — which makes it an instrument, and takes us to instrumental variables.

# References

- Almeida, H., V. Fos and M. Kronlund (2016). The Real Effects of Share Repurchases. *Journal of Financial Economics* 119, 168–185.
- Almond, D., J. Doyle, A. Kowalski and H. Williams (2010). Estimating Marginal Returns to Medical Care. *QJE* 125, 591–634.
- Angrist, J. and V. Lavy (1999). Using Maimonides' Rule to Estimate the Effect of Class Size. *QJE* 114, 533–575.
- Armstrong, T. and M. Kolesár (2020). Simple and Honest Confidence Intervals in Nonparametric Regression. *Quantitative Economics* 11, 1–39.
- Angrist, J. and M. Rokkanen (2015). Wanna Get Away? RD Estimation Away from the Cutoff. *JASA* 110, 1331–1344.
- Barreca, A., M. Guldi, J. Lindo and G. Waddell (2011). Saving Babies? *QJE* 126, 2117–2123.
- Becker, G. (1968). Crime and Punishment: An Economic Approach. *JPE* 76, 169–217.
- Black, S. (1999). Do Better Schools Matter? *QJE* 114, 577–599.
- Brodeur, A., N. Cook and A. Heyes (2020). Methods Matter. *AER* 110, 3634–3660.
- Calonico, S., M. Cattaneo and R. Titiunik (2014). Robust Nonparametric Confidence Intervals for RD Designs. *Econometrica* 82, 2295–2326.
- Card, D., C. Dobkin and N. Maestas (2008). The Impact of Nearly Universal Insurance Coverage on Health Care Utilization. *AER* 98, 2242–2258.
- Carpenter, C. and C. Dobkin (2009). The Effect of Alcohol Consumption on Mortality. *AEJ: Applied* 1, 164–182.
- Cattaneo, M., R. Crump, M. Farrell and Y. Feng (2024). On Binscatter. *AER* 114, 1488–1514.
- Cattaneo, M., B. Frandsen and R. Titiunik (2015). Randomization Inference in the RD Design. *Journal of Causal Inference* 3, 1–24.
- Cattaneo, M., N. Idrobo and R. Titiunik (2020). *A Practical Introduction to Regression Discontinuity Designs*. Cambridge University Press.
- Cattaneo, M., M. Jansson and X. Ma (2020). Simple Local Polynomial Density Estimators. *JASA* 115, 1449–1455.
- Caughey, D. and J. Sekhon (2011). Elections and the Regression Discontinuity Design. *Political Analysis* 19, 385–408.
- Cheng, M.-Y., J. Fan and J. Marron (1997). On Automatic Boundary Corrections. *Annals of Statistics* 25, 1691–1708.
- Cunningham, S. *Causal Inference: The Remix*, Ch. 6 — this lecture follows it closely.
- Cuñat, V., M. Gin'e and M. Guadalupe (2012). The Vote Is Cast. *Journal of Finance* 67, 1943–1977.
- Drago, F., R. Galbiati and P. Vertova (2009). The Deterrent Effects of Prison. *JPE* 117, 257–280.

# References (continued)

- Eggers, A., A. Fowler, J. Hainmueller, A. Hall and J. Snyder (2015). On the Validity of the RD Design for Electoral Effects. *AJPS* 59, 259–274.
- Gelman, A. and G. Imbens (2019). Why High-Order Polynomials Should Not Be Used in RD Designs. *JBES* 37, 447–456.
- Goldberger, A. (1972). Selection Bias in Evaluating Treatment Effects. Discussion paper, Wisconsin.
- Hahn, J., P. Todd and W. van der Klaauw (2001). Identification and Estimation with a RD Design. *Econometrica* 69, 201–209.
- Hansen, B. (2015). Punishment and Deterrence: Evidence from Drunk Driving. *AER* 105, 1581–1617.
- Hoekstra, M. (2009). The Effect of Attending the Flagship State University on Earnings. *ReStat* 91, 717–724.
- Imbens, G. and K. Kalyanaraman (2012). Optimal Bandwidth Choice for the RD Estimator. *ReStud* 79, 933–959.
- Imbens, G. and T. Lemieux (2008). Regression Discontinuity Designs: A Guide to Practice. *Journal of Econometrics* 142, 615–635.
- Keys, B., T. Mukherjee, A. Seru and V. Vig (2010). Did Securitization Lead to Lax Screening? *QJE* 125, 307–362.
- Kolesár, M. and C. Rothe (2018). Inference in RD Designs with a Discrete Running Variable. *AER* 108, 2277–2304.
- Lee, D. (2008). Randomized Experiments from Non-random Selection in U.S. House Elections. *Journal of Econometrics* 142, 675–697.
- Lee, D. and D. Card (2008). Regression Discontinuity Inference with Specification Error. *Journal of Econometrics* 142, 655–674.
- Lee, D. and J. McCrary (2017). The Deterrence Effect of Prison. *Advances in Econometrics* 38, 73–146.
- Lee, D. and T. Lemieux (2010). Regression Discontinuity Designs in Economics. *JEL* 48, 281–355.
- Lee, D., E. Moretti and M. Butler (2004). Do Voters Affect or Elect Policies? *QJE* 119, 807–859.
- Malenko, N. and Y. Shen (2016). The Role of Proxy Advisory Firms. *RFS* 29, 3394–3427.
- McCrary, J. (2008). Manipulation of the Running Variable in the RD Design. *Journal of Econometrics* 142, 698–714.
- Thistlethwaite, D. and D. Campbell (1960). Regression-Discontinuity Analysis. *Journal of Educational Psychology* 51, 309–317.